Comprehensive Notes: What is Feature Engineering?

Course: 100 Days of Machine Learning (Day 23)

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1. Introduction to Feature Engineering

Feature Engineering is considered one of the most critical steps in the Machine Learning pipeline. While the previous sessions focused on Introduction, Data Gathering, and EDA (Exploratory Data Analysis), this session marks the beginning of a deep dive into preparing data for models.

What is Feature Engineering? (Formal Definition)

According to Wikipedia:

"Feature engineering is the process of using domain knowledge to extract features from raw data which can be used to improve the performance of machine learning algorithms."

Simple Explanation

In simple terms, raw data is rarely ready to be fed directly into a machine learning model. You must "prepare" or "cook" the data so the model can understand it better. When you transform raw data into a format that a machine learning model can consume effectively, that process is called **Feature Engineering**.

The "Art" of Machine Learning

The teacher emphasizes that Feature Engineering is more of an **Art** than a hard science.

- Unlike programming where algorithms follow fixed rules, Feature Engineering depends on the **Data Scientist's intuition, domain knowledge, and experience**.
- Different data scientists might approach the same dataset with different feature engineering strategies.

Importance: Feature vs. Algorithm

There is a famous saying in Machine Learning:

"A mediocre algorithm with great features will outperform a great algorithm with mediocre features."

Why? Because the quality of the input (features/columns) directly determines the quality of the output. This is why the course dedicates 10-15 videos to this topic alone.

2. Feature Engineering in the ML Pipeline

In the Machine Learning life cycle, Feature Engineering happens right after initial data pre-processing and before model training. It usually consists of:

1. Feature Transformation
2. Feature Construction
3. Feature Selection

4. Feature Extraction

3. The Four Pillars of Feature Engineering (Flowchart)

I. Feature Transformation

This involves taking an existing feature (column) and transforming it into a different form to make it more suitable for the algorithm.

1. Missing Value Imputation

Real-world data often has missing values due to collection errors or user omissions.

- **Problem:** Libraries like Scikit-Learn generally do not accept missing values.
- **Solution:** You must either remove the rows with missing values or fill them.
- **Techniques:** Filling with Mean, Median, Mode, or other advanced imputation techniques.

2. Handling Categorical Features

Machine learning models only understand numbers (math). If your data has text (e.g., "Dog", "Cat"), you must convert it.

- **Example:** One-Hot Encoding.

Logic Transformation:

Raw Data Column: Animal

Values: ["Dog", "Cat", "Sheep"]

Transformed Data (One-Hot Encoding):

Dog | Cat | Sheep

1 | 0 | 0 -> Representing 'Dog'

0 | 1 | 0 -> Representing 'Cat'

- **Line-by-line logic:**
    - Create a new column for every unique category.
    - If the row originally contained "Dog", put a 1 in the Dog column and 0 in others.
    - This converts text labels into a numerical grid that math models can process.

3. Outlier Detection

Outliers are data points that are significantly different from the rest of the data (e.g., a student scoring 999/100 while everyone else scores around 50).

- **Impact:** Algorithms like Linear Regression are very sensitive to outliers. They "pull" the best-fit line toward the outlier, ruining the model.
- **Goal:** Detect and handle these points before training.

4. Feature Scaling

When features have different ranges (e.g., Age 0-100 and Salary 0-100,000), distance-based algorithms (like KNN) get confused.

- **Reason:** The larger scale (Salary) will dominate the distance calculation, making Age irrelevant.
- **Solution:** Scale both to a similar range (e.g., 0 to 1 or -1 to 1).

II. Feature Construction

This is the process of creating **entirely new features** from existing ones based on domain knowledge or intuition.

Example: Titanic Dataset The dataset has two columns: SibSp (Siblings/Spouses) and Parch (Parents/Children). A data scientist might realize that what actually matters is the **Total Family Size**.

Logic Implementation:

```
# Creating a new feature 'Family_Size'

df['Family_Size'] = df['SibSp'] + df['Parch'] + 1


# Further Categorization:

# If Family_Size == 1 -> "Alone"

# If Family_Size > 1 and < 4 -> "Small Family"

# If Family_Size > 4 -> "Large Family"
```

- **Line-by-line explanation:**
    - df['SibSp'] + df['Parch'] + 1: We sum the relatives and add 1 (for the passenger themselves).
    - This creates one powerful feature instead of two weaker ones.
    - We then group these numbers into categories like "Alone" or "Small Family" to help the model find patterns more easily.

III. Feature Selection

When you have too many features (columns), you select only the most important ones and drop the rest.

Example: MNIST Dataset (Handwritten Digits)

- The images are 28x28 pixels = 784 pixels (features).
- Each pixel is a column in the data table.
- **Teacher's Observation:** Most pixels on the edges of the image are always white/empty. The actual "digit" is only in the center.

- **Goal:** Use Feature Selection to keep only the "useful" center pixels and drop the edge pixels.
- **Benefit:** Improves model speed and performance by reducing noise.

IV. Feature Extraction

Unlike construction (manual) or selection (picking), Extraction uses algorithms to create a **reduced set of new features** that capture the most information from the original data.

Example: Real Estate Instead of having "Number of Rooms" and "Number of Bathrooms," an algorithm might combine them into a single feature called "Square Footage" or a custom "Living Quality Index."

Key Algorithms mentioned:

- **PCA (Principal Component Analysis):** Rotates the data axes to find the directions of maximum variance.
- **LDA (Linear Discriminant Analysis).**

4. Key Takeaways & Summary

- **Feature Engineering is Mandatory:** You cannot build a high-performing model without it.
- **Domain Knowledge Matters:** Knowing the "Why" behind the data helps in creating better features (Feature Construction).
- **Computational Efficiency:** Techniques like Selection and Extraction help handle "High Dimensional Data" (data with too many columns).
- **Roadmap:** The next 10-15 videos will cover each technique (Imputation, Encoding, Scaling, PCA, etc.) in mathematical and programmatic detail.

Tip from Teacher: Don't get overwhelmed by the terminology (PCA, LDA, Imputation). We will treat each one as a separate topic with its own dedicated video.